

DCS DATA FORM — Logic Sim (CFB power plant, ABC drawings)

Unit / DCS: _____ Engineer: _____ Date: _____ Form made for: logic-sim, build after v1.15.1

How to use. Every row gives the TAG, the DESCRIPTION and the ADDRESSES (function block MDL address as written on the drawing, and the field signal address with its IO-list tag) so you can find it in the DCS. The grey boxes are what the simulator uses now (ASSUMED, not from the DCS). Write the real DCS value in the white boxes. A "-" means not used / not present. Order: G single questions and H drum level tables (page 1) · A alarm limits (104) · B PID tuning (68) · C ramp rates (8) · D pulse output (2) · E select circuits (26) · F totalizers (10).

Priority if you have little time: G1–G6 and section H first (a few minutes), then B (PID tuning), then A (alarms). C, D, E, F are short.

Signal address = the AI address read from the drawing (and the IO-list tag/description of that address). Where a PID has several field signals upstream (ratio, 3-element control) all are listed; the first one is the measurement of the loop tag. "no AI on this sheet" means the input is calculated or from an operator station / another sheet (the MDL address is then the way to find it).

G. SINGLE QUESTIONS

#	Where	Question	Sim now	Answer (write / tick)
G1	ABC-001C, timer TR0228 at (212, 162)	The drawing shows a TON of 2 s; the memory list says TP (pulse) 2 s. What is TR0228 in the DCS?	TON 2 s (as drawn)	☐ TON ☐ TP (pulse) time ____ s
G2	PID tuning (section B)	Unit of Ti (integral time)	seconds	☐ sec ☐ min ☐ repeats / min
G3	PID tuning (section B)	Kp is written as	gain (output % per input %)	☐ gain ☐ proportional band %
G4	PID tuning (section B)	Unit of Td (derivative time)	seconds	☐ sec ☐ min
G5	Alarms (section A)	Are the limits in engineering units or in % of range?	engineering units	☐ eng. units ☐ % of range
G6	Alarms (section A)	Deadband / delay of the alarms (if any)	none	deadband ____ delay ____ s

H. DRUM LEVEL COMPENSATION TABLES (ABC-010 F(X) LN38 and LN39)

In the DCS open the linearization tables S1-LN38 and S1-LN39 (function block numbers S1-LN-38 / S1-LN-39). For each drum pressure write the Y value (output) the DCS table gives. The simulator takes the points of the file "Drum Level Calculation.xls": LN38 y = N × 100 (%), LN39 y = K (ratio). If your DCS values are about 100 times different, tell the unit.

Drum pressure (kg/cm2)	LN38 sim now (%)	LN38 DCS (write)	LN39 sim now (ratio)	LN39 DCS (write)
0	100.000		1.0000	
50	97.637		0.7846	
100	94.783		0.6627	
145	91.299		0.5483	
200	81.227		0.2966	
250	77.151		0.1976	

A. ALARM LIMITS (ALM blocks, 104) — read the limits in the DCS / HMI alarm database

Write the limit in the SAME unit as the signal (column Unit). If a limit is not used in the DCS write "-". "Sim now" = what the simulator uses at the moment (ASSUMED by the engineer assistant, NOT from the DCS).

#	Sheet	Tag	Function block (MDL)	Signal address [IO tag]	Description	Unit	Sim now (ASSUMED)				DCS value (write)			
							HH	H	L	LL	HH	H	L	LL
1	ABC-003B	FIFA1043B	S1-MDL169	AI0290 [FT-FA1043-B]	Start-up burner B secondary air flow	T/H	58	50	20	12				
2	ABC-003C	FIFA1043C	S1-MDL170	AI0403 [FT-FA1043-C]	Start-up burner C secondary air flow	T/H	58	50	20	12				
3	ABC-003D	FIFA1043D	S1-MDL171	AI0597 [FT-FA1043-D]	Start-up burner D secondary air flow	T/H	58	50	20	12				
4	ABC-005	SIM1311	S1-MDL104	AI0151 [ST-M1311]	Limestone rotary feeder speed (0-100 %)	%	98	90	5	1				
5	ABC-006	FIFA1081	S1-MDL165	AI0513 [FT-FA1081] AI0628 [TT-FA1082]	SA fan inlet air flow (0-432 T/H)	T/H	410	380	80	40				
6	ABC-006	FIFA1071	S1-MDL166	AI0484 [FT-FA1071] AI0580 [TT-FA1072]	PA fan inlet air flow (0-648 T/H)	T/H	610	560	120	60				
7	ABC-006	FIFA1055	S1-MDL167	AI0545 [FT-FA1055] AI0579 [TT-FA1055-A] AI0627 [TT-FA1055-B]	FA blower outlet common air flow (0-35 T/H)	T/H	33	30	25	20				
8	ABC-006	TIFA1082	S1-MDL221	AI0628 [TT-FA1082]	SA fan inlet air temperature	°C	55	45	5	0				
9	ABC-006	TAF	S1-MDL191	AI0628 [TT-FA1082] AI0513 [FT-FA1081] AI0580 [TT-FA1072] AI0484 [FT-FA1071] AI0579 [TT-FA1055-A] AI0627 [TT-FA1055-B] AI0545 [FT-FA1055]	Total air flow (sum of the air flows, ABC-006)	T/H	900	800	250	150				
10	ABC-006	TIFA1055A	S1-MDL138	AI0579 [TT-FA1055-A]	FA blower A outlet air temperature	°C	130	110	40	20				
11	ABC-006	TIFA1055B	S1-MDL220	AI0627 [TT-FA1055-B]	FA blower B outlet air temperature	°C	130	110	40	20				
12	ABC-006	TIFA1072	S1-MDL215	AI0580 [TT-FA1072]	PA fan inlet air temperature	°C	55	45	5	0				
13	ABC-006	TIFA1076	S1-MDL109	AI0656 [TT-FA1076]	Air preheater outlet to primary air temperature (A)	°C	340	320	150	100				
14	ABC-006	TIFA1077	S1-MDL111	AI0672 [TT-FA1077]	Air preheater outlet to primary air temperature (B)	°C	340	320	150	100				
15	ABC-006	FIFA1079	S1-MDL214	AI0166 [FT-FA1079] AI0656 [TT-FA1076] AI0672 [TT-FA1077]	Air preheater outlet to furnace primary air flow (0-510 T/H)	T/H	480	440	150	100				
16	ABC-008	PIFG1085	S1-MDL085	AI0132 [PT-FG1085]	Furnace pressure A (-600..600)	mmH2O	300	100	-250	-400				
17	ABC-008	PIFG1086	S1-MDL086	AI0260 [PT-FG1086]	Furnace pressure B	mmH2O	300	100	-250	-400				
18	ABC-008	PIFG10871	S1-MDL087	AI0388 [PT-FG1087-1]	Furnace pressure C-1	mmH2O	300	100	-250	-400				
19	ABC-008	SILO1756	S1-MDL154	AI0481 [ST-LO1756]	IDF hydraulic coupling speed (0-1200 rpm)	rpm	1160	1100	200	100				

#	Sheet	Tag	Function block (MDL)	Signal address [IO tag]	Description	Unit	Sim now (ASSUMED)				DCS value (write)			
							HH	H	L	LL	HH	H	L	LL
20	ABC-009A	DPIFG1081	S1-MDL100	AI0134 [DPT-FG1081]	Furnace plenum / upper layer differential pressure A (0-2500)	mmH2O	2200	1900	1150	900				
21	ABC-009A	DPIFG1082	S1-MDL101	AI0262 [DPT-FG1082]	Furnace plenum / upper layer differential pressure B	mmH2O	2200	1900	1150	900				
22	ABC-010	LIBR10012	S1-MDL077	AI0258 [LT-BR1001-2] AI0385 [PT-BR1001-2] AI0128 [PT-BR1001-1]	Drum level (2)	mm	150	75	-75	-150				
23	ABC-010	LIBR10011	S1-MDL075	AI0130 [LT-BR1001-1] AI0128 [PT-BR1001-1] AI0385 [PT-BR1001-2]	Drum level (1) (-422..820 mm)	mm	150	75	-75	-150				
24	ABC-010	LIBR10013	S1-MDL078	AI0386 [LT-BR1001-3] AI0128 [PT-BR1001-1] AI0385 [PT-BR1001-2]	Drum level (3)	mm	150	75	-75	-150				
25	ABC-010	TIFW1160	S1-MDL136	AI0592 [TT-FW1160]	Economiser No.1 inlet water temperature (0-300)	°C	285	270	150	120				
26	ABC-010	PIBR10011	S1-MDL080	AI0128 [PT-BR1001-1]	Drum pressure (1) (0-250)	kg/cm2	168	158	110	90				
27	ABC-010	PIBR10012	S1-MDL081	AI0385 [PT-BR1001-2]	Drum pressure (2)	kg/cm2	168	158	110	90				
28	ABC-011	TIBR11401	S1-MDL130	AI0610 [TT-BR1140-1]	Superheater No.2 outlet steam temperature (1)	°C	550	530	430	400				
29	ABC-011	TIBR11402	S1-MDL131	AI0642 [TT-BR1140-2]	Superheater No.2 outlet steam temperature (2)	°C	550	530	430	400				
30	ABC-011	TIBR11301	S1-MDL128	AI0609 [TT-BR1130-1]	Superheater No.2 inlet steam temperature (1)	°C	490	470	350	320				
31	ABC-011	TIBR11302	S1-MDL129	AI0641 [TT-BR1130-2]	Superheater No.2 inlet steam temperature (2)	°C	490	470	350	320				
32	ABC-012	TIMS10041	S1-MDL134	AI0612 [TT-MS1004-1]	Main steam temperature (1), finishing S/H outlet	°C	560	550	520	500				
33	ABC-012	TIMS10042	S1-MDL135	AI0644 [TT-MS1004-2]	Main steam temperature (2)	°C	560	550	520	500				
34	ABC-012	TIBR11501	S1-MDL132	AI0611 [TT-BR1150-1]	Finishing superheater inlet steam temperature (1)	°C	540	520	440	420				
35	ABC-012	TIBR11502	S1-MDL133	AI0643 [TT-BR1150-2]	Finishing superheater inlet steam temperature (2)	°C	540	520	440	420				
36	ABC-019	TIHR10021	S2-MDL120	AI0867 [TT-HR1002-1]	Reheater outlet header temperature (1)	°C	560	550	515	500				
37	ABC-019	TIHR10022	S2-MDL121	AI0899 [TT-HR1002-2]	Reheater outlet header temperature (2)	°C	560	550	515	500				
38	ABC-019	TICR10051	S2-MDL118	AI0866 [TT-CR1005-1]	Desuperheater temperature (1), reheater inlet	°C	400	380	280	250				
39	ABC-019	TICR10052	S2-MDL119	AI0898 [TT-CR1005-2]	Desuperheater temperature (2), reheater inlet	°C	400	380	280	250				

#	Sheet	Tag	Function block (MDL)	Signal address [IO tag]	Description	Unit	Sim now (ASSUMED)				DCS value (write)			
							HH	H	L	LL	HH	H	L	LL
40	ABC-019	PICR1006	S2-MDL063	AI0529 [PT-CR1006]	Reheater inlet header pressure (0-50)	kg/cm2	36	33	5	2				
41	ABC-026	TIFG1091A	S3-MDL107	AI0816 [TT-FG1091-A]	Furnace lower layer ABOVE temperature A (0-1200)	°C	1020	960	800	700				
42	ABC-026	TIFG1091B	S3-MDL108	AI0817 [TT-FG1091-B]	ABOVE temperature B	°C	1020	960	800	700				
43	ABC-026	TIFG1091C	S3-MDL109	AI0818 [TT-FG1091-C]	ABOVE temperature C	°C	1020	960	800	700				
44	ABC-026	TIFG1091F	S3-MDL112	AI0821 [TT-FG1091-F]	ABOVE temperature F	°C	1020	960	800	700				
45	ABC-026	TIFG1091G	S3-MDL113	AI0848 [TT-FG1091-G]	ABOVE temperature G	°C	1020	960	800	700				
46	ABC-026	TIFG1091	S3-MDL237	AI0820 [TT-FG1091-E] AI0819 [TT-FG1091-D] AI0818 [TT-FG1091-C] AI0817 [TT-FG1091-B] AI0816 [TT-FG1091-A] AI0821 [TT-FG1091-F] AI0848 [TT-FG1091-G] AI0849 [TT-FG1091-H] AI0850 [TT-FG1091-I]	Furnace total average temperature	°C	1020	960	800	700				
47	ABC-026	TIFG1091D	S3-MDL110	AI0819 [TT-FG1091-D]	ABOVE temperature D	°C	1020	960	800	700				
48	ABC-026	TIFG1091E	S3-MDL111	AI0820 [TT-FG1091-E]	ABOVE temperature E	°C	1020	960	800	700				
49	ABC-026	TIFG1091H	S3-MDL114	AI0849 [TT-FG1091-H]	ABOVE temperature H	°C	1020	960	800	700				
50	ABC-026	TIFG1091I	S3-MDL115	AI0850 [TT-FG1091-I]	ABOVE temperature I	°C	1020	960	800	700				
51	ABC-027	TIFG1090A	S3-MDL193	AI0832 [TT-FG1090-A]	Furnace lower layer UNDER temperature A (0-1200)	°C	1000	950	750	650				
52	ABC-027	TIFG1090B	S3-MDL194	AI0833 [TT-FG1090-B]	UNDER temperature B	°C	1000	950	750	650				
53	ABC-027	TIFG1090C	S3-MDL195	AI0834 [TT-FG1090-C]	UNDER temperature C	°C	1000	950	750	650				
54	ABC-027	TIFG1090F	S3-MDL198	AI0837 [TT-FG1090-F]	UNDER temperature F	°C	1000	950	750	650				
55	ABC-027	TIFG1090G	S3-MDL199	AI0864 [TT-FG1090-G]	UNDER temperature G	°C	1000	950	750	650				
56	ABC-027	TIFG1090D	S3-MDL196	AI0835 [TT-FG1090-D]	UNDER temperature D	°C	1000	950	750	650				
57	ABC-027	TIFG1090E	S3-MDL197	AI0836 [TT-FG1090-E]	UNDER temperature E	°C	1000	950	750	650				
58	ABC-027	TIFG1090H	S3-MDL200	AI0865 [TT-FG1090-H]	UNDER temperature H	°C	1000	950	750	650				

#	Sheet	Tag	Function block (MDL)	Signal address [IO tag]	Description	Unit	Sim now (ASSUMED)				DCS value (write)			
							HH	H	L	LL	HH	H	L	LL
59	ABC-027	TIFG1090I	S3-MDL201	AI0866 [TT-FG1090-I]	UNDER temperature I	°C	1000	950	750	650				
60	ABC-027	TIFG1090L	S3-MDL239	AI0836 [TT-FG1090-E] AI0835 [TT-FG1090-D] AI0834 [TT-FG1090-C] AI0833 [TT-FG1090-B] AI0832 [TT-FG1090-A] AI0837 [TT-FG1090-F] AI0864 [TT-FG1090-G] AI0865 [TT-FG1090-H] AI0866 [TT-FG1090-I]	Furnace lower layer under temperature average	°C	1000	950	750	650				
61	ABC-030	PIAS1004	S2-MDL056	AI0514 [PT-AS1004]	Auxiliary steam supply pressure, main steam pipe (0-15)	kg/cm2	13.5	12	6	4				
62	ABC-030	PIAS1005	S2-MDL059	AI0515 [PT-AS1005]	Auxiliary steam supply pressure, cold reheat	kg/cm2	13.5	12	6	4				
63	ABC-031	LICD11041	S2-MDL033	AI0672 [LT-CD1104-1]	Deaerator storage tank level (1) (-2335..1095)	mm	700	400	-400	-800				
64	ABC-031	LICD11042	S2-MDL034	AI0848 [LT-CD1104-2]	Deaerator storage tank level (2)	mm	700	400	-400	-800				
65	ABC-031	LICD11043	S2-MDL035	AI0928 [LT-CD1104-3]	Deaerator storage tank level (3)	mm	700	400	-400	-800				
66	ABC-035	LIHD11052	S2-MDL079	AI0852 [LT-HD1105-2]	No.1 HP heater level (2)	mm	200	100	-100	-200				
67	ABC-035	LIHD11051	S2-MDL078	AI0676 [LT-HD1105-1]	No.1 HP heater level (1) (-300..310)	mm	200	100	-100	-200				
68	ABC-036	LIHD11061	S2-MDL080	AI0677 [LT-HD1106-1]	No.2 HP heater level (1)	mm	200	100	-100	-200				
69	ABC-036	LIHD11062	S2-MDL081	AI0853 [LT-HD1106-2]	No.2 HP heater level (2)	mm	200	100	-100	-200				
70	ABC-037	LIHD11011	S2-MDL072	AI0673 [LT-HD1101-1]	No.1 LP heater level (1)	mm	200	100	-100	-200				
71	ABC-037	LIHD11012	S2-MDL073	AI0849 [LT-HD1101-2]	No.1 LP heater level (2)	mm	200	100	-100	-200				
72	ABC-038	LIHD11021	S2-MDL074	AI0674 [LT-HD1102-1]	No.2 LP heater level (1)	mm	200	100	-100	-200				
73	ABC-038	LIHD11022	S2-MDL075	AI0850 [LT-HD1102-2]	No.2 LP heater level (2)	mm	200	100	-100	-200				
74	ABC-039	LIHD11031	S2-MDL076	AI0675 [LT-HD1103-1]	No.3 LP heater level (1)	mm	200	100	-100	-200				
75	ABC-039	LIHD11032	S2-MDL077	AI0851 [LT-HD1103-2]	No.3 LP heater level (2)	mm	200	100	-100	-200				
76	ABC-050	PIMS10061	S2-MDL036	AI0416 [PT-MS1006-1]	Turbine main steam pressure (1) (0-150)	kg/cm2	147	143	110	90				
77	ABC-050	PIMS10062	S2-MDL037	AI0512 [PT-MS1006-2]	Turbine main steam pressure (2)	kg/cm2	147	143	110	90				

#	Sheet	Tag	Function block (MDL)	Signal address [IO tag]	Description	Unit	Sim now (ASSUMED)				DCS value (write)			
							HH	H	L	LL	HH	H	L	LL
78	ABC-051	TIMS10221	S2-MDL116	AI0865 [TT-MS1022-1]	HP turbine bypass outlet temperature (1)	°C	400	380	250	200				
79	ABC-051	TIMS10222	S2-MDL117	AI0897 [TT-MS1022-2]	HP turbine bypass outlet temperature (2)	°C	400	380	250	200				
80	ABC-051	PICR1003	S2-MDL060	AI0528 [PT-CR1003]	Cold reheat steam pressure	kg/cm2	36	33	5	2				
81	ABC-051	TICR1004	S2-MDL124	AI0900 [TT-CR1004]	Cold reheat steam temperature	°C	400	380	250	200				
82	ABC-051	PIMS1021	S2-MDL134	AI0566 [PT-MS1021]	HP turbine bypass outlet pressure	kg/cm2	36	33	5	2				
83	ABC-052	PIHR10031	S2-MDL042	AI0421 [PT-HR1003-1]	Hot reheat steam pressure (1)	kg/cm2	34	31	5	2				
84	ABC-052	PIHR10032	S2-MDL043	AI0513 [PT-HR1003-2]	Hot reheat steam pressure (2)	kg/cm2	34	31	5	2				
85	ABC-052	PIST10011	S2-MDL191	AI0883 [PT-ST1001-1]	HP turbine first stage pressure A (0-150)	kg/cm2	120	110	5	1				
86	ABC-052	PIST10012	S2-MDL192	AI0884 [PT-ST1001-2]	HP turbine first stage pressure B	kg/cm2	120	110	5	1				
87	ABC-052	PIST10013	S2-MDL193	AI0885 [PT-ST1001-3]	HP turbine first stage pressure C	kg/cm2	120	110	5	1				
88	ABC-052	TIHR10121	S2-MDL114	AI0864 [TT-HR1012-1]	LP turbine bypass outlet temperature (1) (0-300)	°C	250	220	100	70				
89	ABC-052	TIHR10122	S2-MDL115	AI0896 [TT-HR1012-2]	LP turbine bypass outlet temperature (2)	°C	250	220	100	70				
90	ABC-052	TIHR1004	S2-MDL122	AI0868 [TT-HR1004]	Hot reheat steam temperature	°C	560	550	515	500				
91	ABC-053	TILO10021	S2-MDL138	AI0452 [TE-LO1002-1]	Turbine oil cooler outlet oil temperature A	°C	55	48	30	25				
92	ABC-053	TILO10022	S2-MDL139	AI0516 [TE-LO1002-2]	Turbine oil cooler outlet oil temperature B	°C	55	48	30	25				
93	ABC-053	TCSAI1004	S2-MDL141	AI0534 [TCS-AI1004]	Turbine rated speed (0-4000 rpm)	rpm	3300	3060	2950	2800				
94	ABC-054	ZIHR1391	S4-MDL101	AI0737 [ZT-HR1391]	Blow-off M.V. position, hot reheat steam (0-100 %)	%	50	10						
95	ABC-057	PIFA1085	S1-MDL093	AI0288 [PT-FA1085]	SAF outlet to air preheater air pressure (0-3000)	mmH2O	2700	2400	800	500				
96	ABC-001C	GCPAI1001	S1-MDL192	AI0419 [GCP-AI1001]	Unit MW actual C (0-240)	MW	170	160	45	30				
97	ABC-001C	GCPAI1002	S1-MDL193	AI0774 [GCP-AI1002]	Unit MW actual D (0-240)	MW	170	160	45	30				

#	Sheet	Tag	Function block (MDL)	Signal address [IO tag]	Description	Unit	Sim now (ASSUMED)				DCS value (write)			
							HH	H	L	LL	HH	H	L	LL
98	ABC-001C	TCSAI1001	S1-MDL226	AI0420 [TCS-AI1001]	Target MW set point from TCS (0-200)	MW	170	160	45	30				
99	ABC-002	AIFG1122	S1-MDL144	AI0273 [AT-FG1122]	Flue gas O2 A (0-25 %)	%	8	6	2	1				
100	ABC-002	AIFG1123	S1-MDL145	AI0433 [AT-FG1123]	Flue gas O2 B	%	8	6	2	1				
101	ABC-003A	TIFA1086	S1-MDL114	AI0657 [TT-FA1086]	Secondary air (hot) temperature A	°C	350	330	150	100				
102	ABC-003A	TIFA1087	S1-MDL115	AI0673 [TT-FA1087]	Secondary air (hot) temperature B	°C	350	330	150	100				
103	ABC-003A	FIFA1043A	S1-MDL168	AI0164 [FT-FA1043-A] AI0657 [TT-FA1086] AI0673 [TT-FA1087]	Start-up burner A secondary air flow (0-65 T/H)	T/H	58	50	20	12				
104	ABC-003A	PIFA1088	S1-MDL094	AI0289 [PT-FA1088]	Air preheater to furnace secondary air pressure (0-3000)	mmH2O	2700	2400	1000	900				

B. PID / PIDV TUNING (68 controllers) — read from the controller tuning / faceplate in the DCS

Answer G2, G3 and G4 (same page as the drum level tables) first: unit of Ti, type of Kp, unit of Td. Dir: D = direct (output rises when PV rises), R = reverse. Out lo / hi = output limits in %. "Sim now" = default tuning of the simulator (by loop type, NOT from the DCS).

#	Sheet	Kind	Tag	Function block (MDL)	PV signal address [IO tag]	Description	Range / unit	Sim now (ASSUMED)				DCS value (write)					
								Kp	Ti	Td	Dir	Kp	Ti	Td	Dir	Out lo	Out hi
1	ABC-003B	PID	FICFA1043B	S1-MDL028	AI0290 [FT-FA1043-B] AI0546 [FT-DO1043-B]	#1 Boiler start up burner-B hot secondary air inlet flow	0 ~ 65 T/H	1	15	0	R						
2	ABC-003B	PID	PICDO1043B	S1-MDL063	AI0439 [PT-DO1043-B2]	#1 Boiler start up burner-B diesel oil supply line oil pressure	kg/cm2	1	60	0	R						
3	ABC-003B	PID	FICDO1043B	S1-MDL034	AI0546 [FT-DO1043-B]	#1 Boiler start up burner-B supply line oil flow	0 ~ 3.5 T/H	1	15	0	R						
4	ABC-003C	PID	FICFA1043C	S1-MDL197	AI0403 [FT-FA1043-C] AI0517 [FT-DO1043-C]	#1 Boiler start up burner-C hot secondary air inlet flow	0 ~ 65 T/H	1	15	0	R						
5	ABC-003C	PID	PICDO1043C	S1-MDL064	AI0596 [PT-DO1043-C2]	#1 Boiler start up burner-C diesel oil supply line oil pressure	kg/cm2	1	60	0	R						
6	ABC-003C	PID	FICDO1043C	S1-MDL035	AI0517 [FT-DO1043-C]	#1 Boiler start up burner-C supply line oil flow	0 ~ 3.5 T/H	1	15	0	R						
7	ABC-003D	PID	FICFA1043D	S1-MDL198	AI0597 [FT-FA1043-D] AI0548 [FT-DO1043-D]	#1 Boiler start up burner-D hot secondary air inlet flow	0 ~ 65 T/H	1	15	0	R						
8	ABC-003D	PID	PICDO1043D	S1-MDL065	AI0771 [PT-DO1043-D2]	#1 Boiler start up burner-D diesel oil supply line oil pressure	kg/cm2	1	60	0	R						
9	ABC-003D	PID	FICDO1043D	S1-MDL038	AI0548 [FT-DO1043-D]	#1 Boiler start up burner-D supply line oil flow	0 ~ 3.5 T/H	1	15	0	R						
10	ABC-003E	PID	FIC-OM	S1-MDL164	no AI on this sheet (calculated / operator)	OIL MASTER CONTROL		1	15	0	R						
11	ABC-004A	PID	FIC-CM	S1-MDL015	AI0480 [MIF-CL1061-A]	#1 Coal feeder-A feedrate	0 ~ 120 T/H	1	15	0	R						
12	ABC-004A	PID	FIC-CF	S1-MDL014	AI0480 [MIF-CL1061-A]	#1 Coal feeder-A feedrate	0 ~ 100 %	1	15	0	R						
13	ABC-004A	PID	HS-COAL	S1-MDL051	no AI on this sheet (calculated / operator) - inputs: COS manual value	COAL MASTER CONTROL & A-COAL FEEDER CONTROL		1	60	0	R						
14	ABC-005	PID	AICFG10571	S1-MDL037	AI0466 [AT-FG1057-1]	#1 Stack outlet flue gas SOx value	0 ~ 500 ppm	0.8	240	0	D						
15	ABC-007	PID	FICFA1081	S1-MDL024	signal comes from another sheet - inputs: M.0098, M.025B, COS manual value	PAF/SAF AIR FLOW CONTROL	0 ~ 432 T/H	1	15	0	R						
16	ABC-007	PID	FICFA1071	S1-MDL023	signal comes from another sheet - inputs: COS manual value	PAF/SAF AIR FLOW CONTROL	0 ~ 648 T/H	1	15	0	R						
17	ABC-008	PID	PICFG108	S1-MDL036	no AI on this sheet (calculated / operator) - inputs: SI0180	FURNACE PRESSURE CONTROL	-600 ~ 600 mmH2O	1	60	0	D						
18	ABC-008	PID	PICFG108A	S1-MDL029	no AI on this sheet (calculated / operator) - inputs: SI0180	FURNACE PRESSURE CONTROL	-600 ~ 600 mmH2O	1	60	0	D						

#	Sheet	Kind	Tag	Function block (MDL)	PV signal address [IO tag]	Description	Range / unit	Sim now (ASSUMED)				DCS value (write)					
								Kp	Ti	Td	Dir	Kp	Ti	Td	Dir	Out lo	Out hi
19	ABC-009A	PID	DPICFG108	S1-MDL031	AI0134 [DPT-FG1081] AI0262 [DPT-FG1082]	#1 Furnace plenum/upper layer diff. pressure (A)	0 ~ 2500 mmH2O	1	60	0	D						
20	ABC-009A	PID	PIC-BASC	S1-MDL032	AI0134 [DPT-FG1081] AI0262 [DPT-FG1082]	#1 Furnace plenum/upper layer diff. pressure (A)	0 ~ 100 %	1	60	0	R						
21	ABC-010	PID	LICBR1001	S1-MDL020	AI0130 [LT-BR1001-1] AI0128 [PT-BR1001-1] AI0385 [PT-BR1001-2] AI0386 [LT-BR1001-3]	#1 Boiler drum level (1)	-422 ~ 820 mm	1	120	0	R						
22	ABC-010	PID	LICBR1001A	S1-MDL021	AI0130 [LT-BR1001-1] AI0128 [PT-BR1001-1] AI0385 [PT-BR1001-2] AI0386 [LT-BR1001-3]	#1 Boiler drum level (1)	-422 ~ 820 mm	1	120	0	R						
23	ABC-010	PID	FICFW1007	S1-MDL022	signal comes from another sheet - inputs: SI0269, B.0689, B.0640, M.3344	DRUM LEVEL CONTROL (FEED WATER FLOW CONTROL)	0 ~ 550 T/H	1	15	0	R						
24	ABC-010	PID	LICBR1001B	S1-MDL049	AI0130 [LT-BR1001-1] AI0128 [PT-BR1001-1] AI0385 [PT-BR1001-2] AI0386 [LT-BR1001-3]	#1 Boiler drum level (1)	-422 ~ 820 mm	1	120	0	R						
25	ABC-011	PID	TICBR1130	S1-MDL016	AI0609 [TT-BR1130-1] AI0641 [TT-BR1130-2]	#1 Superheater NO.2 inlet header steam temperature (1)	0 ~ 600 °C	1	180	0	D						
26	ABC-011	PID	TICBR1140	S1-MDL017	AI0610 [TT-BR1140-1] AI0642 [TT-BR1140-2]	#1 Superheater NO.2 outlet header steam temperature (1)	0 ~ 600 °C	1	180	0	R						
27	ABC-012	PID	TICBR1150	S1-MDL018	AI0611 [TT-BR1150-1] AI0643 [TT-BR1150-2]	#1 Finishing S/H inlet header steam temperature (1)	0 ~ 600 °C	1	180	0	D						
28	ABC-012	PID	TICMS1004	S1-MDL009	AI0612 [TT-MS1004-1] AI0644 [TT-MS1004-2]	#1 Boiler finishing S/H outlet main steam temp. (1)	°C	1	180	0	R						
29	ABC-013	PID	PICMS1002	S1-MDL030	no AI on this sheet (calculated / operator)	MAIN STEAM PRESSURE CONTROL (BLOW-OFF)	0 ~ 150 kg/cm2	1	60	0	D						
30	ABC-014	PID	FICFA1055A	S1-MDL025	no AI on this sheet (calculated / operator)	FLUIDIZING AIR FLOW CONTROL	0 ~ 35 T/H	1	15	0	R						
31	ABC-014	PID	FICFA1055B	S1-MDL026	no AI on this sheet (calculated / operator)	FLUIDIZING AIR FLOW CONTROL	0 ~ 35 T/H	1	15	0	R						
32	ABC-017	PID	PICSB1052	S2-MDL001	AI0533 [PT-SB1052]	#1 Boiler sootblower header steam pressure	0 ~ 50 kg/cm2	1	60	0	R						
33	ABC-019	PID	TICHR1002	S2-MDL005	AI0867 [TT-HR1002-1] AI0899 [TT-HR1002-2]	#1 Reheater outlet header temperature (1)	0 ~ 600 °C	1	180	0	D						
34	ABC-020	PID	TICHR1002A	S2-MDL006	no AI on this sheet (calculated / operator) - inputs: HICHR1002A.MAN, TICHR1002A.LOC, COS manual value	REHEATER/SUPERHEATER FLUE GAS DAMPER CONTROL	0 ~ 600 °C	1	180	0	R						
35	ABC-028	PID	PICDO0005	S5-MDL001	AI0181 [PT-DO0005]	Diesel oil pump outlet to start-up burner oil pressure	0 ~ 25 kg/cm2	1	60	0	D						
36	ABC-029	PID	PICDO1008	S1-MDL019	AI0514 [PT-DO1008]	#1 Boiler start up burner diesel oil circulation line oil pressure	0 ~ 25 kg/cm2	1	60	0	D						

#	Sheet	Kind	Tag	Function block (MDL)	PV signal address [IO tag]	Description	Range / unit	Sim now (ASSUMED)				DCS value (write)					
								Kp	Ti	Td	Dir	Kp	Ti	Td	Dir	Out lo	Out hi
37	ABC-030	PID	PICAS1005	S2-MDL008	AI0515 [PT-AS1005] AI0514 [PT-AS1004]	#1 Cold reheat to aux. steam supply pressure	0 ~ 15 kg/cm2	1	60	0	R						
38	ABC-030	PID	PICAS1004	S2-MDL007	AI0514 [PT-AS1004] AI0515 [PT-AS1005]	#1 Main steam pipe to aux. steam supply pressure	0 ~ 15 kg/cm2	1	60	0	R						
39	ABC-031	PID	LICCD1110A	S2-MDL010	AI0640 [LT-CD1110]	#1 Condenser hot well water level	-436 ~ 464 mm	1	120	0	R						
40	ABC-031	PID	LICCD1110B	S2-MDL011	AI0640 [LT-CD1110]	#1 Condenser hot well water level	-436 ~ 464 mm	1	120	0	D						
41	ABC-031	PID	LICCD1104	S2-MDL004	no AI on this sheet (calculated / operator)	COND. DUMPING TO RAW PIT / COND. MAKE-UP WATER / DEAERATOR LEVEL CONTROL	-2335 ~ 1095 mm	1	120	0	R						
42	ABC-032	PID	FICCD1121	S2-MDL013	AI0436 [FT-CD1121] AI0437 [FT-CD1120]	#1 Condensate polisher inlet water flow	0 ~ 200 T/H	1	15	0	D						
43	ABC-032	PID	FICCD1120	S2-MDL012	AI0437 [FT-CD1120]	#1 Condensate water flow	0 ~ 400 T/H	1	15	0	R						
44	ABC-033	PID	TICAS1001	S2-MDL014	AI0641 [TT-AS1001]	#1 Main steam to aux. steam desuperheater temperature	0 ~ 600 °C	1	180	0	D						
45	ABC-034	PID	TICAS1017	S2-MDL015	AI0902 [TT-AS1017]	#1 Gland steam turbine side temperature	0 ~ 300 °C	1	180	0	D						
46	ABC-035	PID	LICHHD11051	S2-MDL016	AI0676 [LT-HD1105-1] AI0852 [LT-HD1105-2]	#1 NO.1 HPH level (1)	-300 ~ 310 mm	1	120	0	D						
47	ABC-035	PID	LICHHD11052	S2-MDL017	AI0852 [LT-HD1105-2] AI0676 [LT-HD1105-1]	#1 NO.1 HPH level (2)	-300 ~ 310 mm	1	120	0	D						
48	ABC-036	PID	LICHHD11061	S2-MDL018	AI0677 [LT-HD1106-1] AI0853 [LT-HD1106-2]	#1 NO.2 HPH level (1)	-300 ~ 310 mm	1	120	0	D						
49	ABC-036	PID	LICHHD11062	S2-MDL019	AI0853 [LT-HD1106-2] AI0677 [LT-HD1106-1]	#1 NO.2 HPH level (2)	-300 ~ 310 mm	1	120	0	D						
50	ABC-037	PID	LICHHD11011	S2-MDL020	AI0673 [LT-HD1101-1] AI0849 [LT-HD1101-2]	#1 NO.1 LPH level (1)	-320 ~ 290 mm	1	120	0	D						
51	ABC-037	PID	LICHHD11012	S2-MDL021	AI0849 [LT-HD1101-2] AI0673 [LT-HD1101-1]	#1 NO.1 LPH level (2)	-320 ~ 290 mm	1	120	0	D						
52	ABC-038	PID	LICHHD11021	S2-MDL022	AI0674 [LT-HD1102-1] AI0850 [LT-HD1102-2]	#1 NO.2 LPH level (1)	-315 ~ 295 mm	1	120	0	D						
53	ABC-038	PID	LICHHD11022	S2-MDL023	AI0850 [LT-HD1102-2] AI0674 [LT-HD1102-1]	#1 NO.2 LPH level (2)	-315 ~ 295 mm	1	120	0	D						
54	ABC-039	PID	LICHHD11031	S2-MDL024	AI0675 [LT-HD1103-1] AI0851 [LT-HD1103-2]	#1 NO.3 LPH level (1)	-310 ~ 300 mm	1	120	0	D						
55	ABC-039	PID	LICHHD11032	S2-MDL025	AI0851 [LT-HD1103-2] AI0675 [LT-HD1103-1]	#1 NO.3 LPH level (2)	-310 ~ 300 mm	1	120	0	D						

#	Sheet	Kind	Tag	Function block (MDL)	PV signal address [IO tag]	Description	Range / unit	Sim now (ASSUMED)				DCS value (write)					
								Kp	Ti	Td	Dir	Kp	Ti	Td	Dir	Out lo	Out hi
56	ABC-050	PID	PICMS1006	S2-MDL002	AI0416 [PT-MS1006-1] AI0512 [PT-MS1006-2]	#1 Turbine main steam pressure(1)	0 ~ 150 kg/cm2	1	60	0	D						
57	ABC-051	PID	TICMS1022	S2-MDL003	AI0865 [TT-MS1022-1] AI0897 [TT-MS1022-2]	#1 HP turbine bypass outlet temperature (1)	0 ~ 600 °C	1	180	0	D						
58	ABC-052	PID	PICHR1003	S2-MDL026	AI0883 [PT-ST1001-1] AI0884 [PT-ST1001-2] AI0885 [PT-ST1001-3] AI0421 [PT-HR1003-1] AI0513 [PT-HR1003-2]	#1 HP turbine first stage steam pressure (A)	0 ~ 50 kg/cm2	1	60	0	D						
59	ABC-052	PID	TICHR1012	S2-MDL027	AI0864 [TT-HR1012-1] AI0896 [TT-HR1012-2]	#1 LP turbine bypass outlet temperature (1)	0 ~ 300 °C	1	180	0	D						
60	ABC-053	PID	TICLO1002	S2-MDL028	AI0534 [TCS-AI1004] AI0452 [TE-LO1002-1] AI0516 [TE-LO1002-2]	#1 Turbine rated speed	0 ~ 100 °C	1	180	0	D						
61	ABC-054	PIDV	PICHR1391	S1-MDL046	AI0737 [ZT-HR1391]	#1 Hot reheat steam blow-off motor valve position	0 ~ 50 kg/cm2	1	60	0	D						
62	ABC-055	PIDV	HICBR1391	S1-MDL071	AI0181 [ZT-BR1391]	#1 S/H NO.1 outlet header to reheater motor valve position	0 ~ 100 %	1	60	0	R						
63	ABC-056	PID	PICCD1104	S2-MDL009	AI0417 [PT-CD1104]	#1 Boiler deaerator pressure	0 ~ 10 kg/cm2	1	60	0	R						
64	ABC-001C	PID	PIC-COBM	S1-MDL006	AI0131 [PT-MS1002]	#1 Boiler finishing superheater outlet main steam pressure	0 ~ 150 Kg/cm2	1	60	0	R						
65	ABC-002	PID	AICFG112	S1-MDL010	signal comes from another sheet - inputs: M.0101, AICFG112.LOC, COS manual value	COMBUSTION AIR CONTROL	0 ~ 25 %	0.8	240	0	R						
66	ABC-003A	PID	FICFA1043A	S1-MDL027	AI0164 [FT-FA1043-A] AI0515 [FT-DO1043-A] AI0657 [TT-FA1086] AI0673 [TT-FA1087]	#1 Boiler start up burner-A hot secondary air inlet flow	0 ~ 65 T/H	1	15	0	R						
67	ABC-003A	PID	PICDO1043A	S1-MDL062	AI0293 [PT-DO1043-A2]	#1 Boiler start up burner-A diesel oil supply line oil pressure	kg/cm2	1	60	0	R						
68	ABC-003A	PID	FICDO1043A	S1-MDL033	AI0515 [FT-DO1043-A]	#1 Boiler start up burner-A supply line oil flow	0 ~ 3.5 T/H	1	15	0	R						

C. RAMP / RATE BOXES with no number on the drawing (8) — read the ramp rate in the DCS

Write the rate and its unit exactly as the DCS shows it (for example 1 T/H per sec, or 2 % per min).

#	Sheet	Loop	Sim now (ASSUMED)	Unit used	DCS rate (write)	DCS unit (write)
1	ABC-004A	FICCL1061A coal feeder A flow control (0-40 T/H), AUTO-MV transfer	1	T/H per s		
2	ABC-004B	FICCL1061B coal feeder B flow control (0-40 T/H)	1	T/H per s		
3	ABC-004C	FICCL1061C coal feeder C flow control (0-40 T/H)	1	T/H per s		
4	ABC-009A	SICL1060A bottom ash screw cooler A speed control (manual mode ramp)	1	% per s		
5	ABC-009A	SICL1060B bottom ash screw cooler B speed control	1	% per s		
6	ABC-009A	SICL1060C bottom ash screw cooler C speed control	1	% per s		
7	ABC-009B	SICL1060D bottom ash screw cooler D speed control	1	% per s		
8	ABC-020	HICHR1002B superheater pass flue gas damper position (manual mode ramp)	2	% per s		

D. PULSE OUTPUT (PO blocks, 2) — valve driven by raise / lower pulses

Cycle = time between pulses, Full stroke = time of the actuator from 0 to 100 %, Shortest pulse = the smallest pulse the DCS sends. Write also the digital output addresses (raise / lower).

#	Sheet	Controller (MDL)	Description	Sim now (ASSUMED)			DCS value (write)				
				Cycle s	Full stroke s	Shortest pulse s	Cycle s	Full stroke s	Shortest s	DO raise address	DO lower address
1	ABC-054	PIDV PICH1391 S1-MDL046	HOT REHEAT STEAM PRESSURE CONTROL (BLOW-OFF)	2	60	0.2					
2	ABC-055	PIDV HICBR1391 S1-MDL071	NO.1 S/H OUTLET HEADER TO REHEATER MOTOR VALVE POSITION CONTROL	2	60	0.2					

E. SELECT CIRCUITS (SEL blocks, 26) — default mode when the operator has not chosen

Write the default mode in the DCS: AVG (average of the healthy inputs), PRI (primary = first input) or SEC (secondary). If the DCS has no choice, write "fixed" and the rule.

#	Sheet	Place (x,y)	Drawing text	Possible modes	Input 1 address [IO tag]	Input 2 address [IO tag]	Sim now	DCS default (write)
1	ABC-006	402,513	AVERAGE TOTAL AIR FLOW (PAF / SAF / FAB)	AVG	AI0656 [TT-FA1076]	AI0672 [TT-FA1077]	AVG	
2	ABC-009A	292,612	PRI /SEC /AVG FURNACE DIFF. PRESSURE CONTROL (A/B/C-BOTTOM ASH SCREW COOLER)	AVG / PRI / SEC	AI0134 [DPT-FG1081]	AI0262 [DPT-FG1082]	AVG	
3	ABC-010	634,486	(1) / (3) / AVG DRUM LEVEL CONTROL (FEED WATER FLOW CONTROL)	AVG / (1) / (3)	AI0130 [LT-BR1001-1] AI0128 [PT-BR1001-1]	AI0386 [LT-BR1001-3] AI0128 [PT-BR1001-1]	AVG	
4	ABC-010	808,601	PRI /SEC /AVG DRUM LEVEL CONTROL (FEED WATER FLOW CONTROL)	AVG / PRI / SEC	AI0128 [PT-BR1001-1]	AI0385 [PT-BR1001-2]	AVG	
5	ABC-011	329,564	PRI /SEC /AVG SUPERHEATER NO.2 OUTLET STEAM TEMP. CONTROL	AVG / PRI / SEC	AI0610 [TT-BR1140-1]	AI0642 [TT-BR1140-2]	AVG	
6	ABC-011	598,564	PRI /SEC /AVG SUPERHEATER NO.2 OUTLET STEAM TEMP. CONTROL	AVG / PRI / SEC	AI0609 [TT-BR1130-1]	AI0641 [TT-BR1130-2]	AVG	
7	ABC-012	358,571	PRI /SEC /AVG MAIN STEAM TEMPERATURE CONTROL	AVG / PRI / SEC	AI0612 [TT-MS1004-1]	AI0644 [TT-MS1004-2]	AVG	
8	ABC-012	635,571	PRI /SEC /AVG MAIN STEAM TEMPERATURE CONTROL	AVG / PRI / SEC	AI0611 [TT-BR1150-1]	AI0643 [TT-BR1150-2]	AVG	
9	ABC-019	434,581	PRI /SEC /AVG REHEATER OUTLET HEADER TEMP. CONTROL	AVG / PRI / SEC	AI0867 [TT-HR1002-1]	AI0899 [TT-HR1002-2]	AVG	
10	ABC-019	634,581	PRI /SEC /AVG REHEATER OUTLET HEADER TEMP. CONTROL	AVG / PRI / SEC	AI0866 [TT-CR1005-1]	AI0898 [TT-CR1005-2]	AVG	
11	ABC-030	576,596	PRI /SEC /AVG MAIN STEAM TO AUX. STEAM PRESSURE / COLD REHEAT TO AUX. STEAM PRESSURE	AVG / PRI / SEC	AI0514 [PT-DO1008]	AI0515 [FT-DO1043-A]	AVG	
12	ABC-035	291,540	PRI /SEC /AVG NO.1 HPH LEVEL CONTROL	AVG / PRI / SEC	AI0676 [TT-FG1096-A]	AI0852 [LT-HD1105-2]	AVG	
13	ABC-036	291,540	PRI /SEC /AVG NO.2 HPH LEVEL CONTROL	AVG / PRI / SEC	AI0677 [TT-FG1096-B]	AI0853 [LT-HD1106-2]	AVG	
14	ABC-037	291,540	PRI /SEC /AVG NO.1 LPH LEVEL CONTROL	AVG / PRI / SEC	AI0673 [TT-FA1087]	AI0849 [LT-HD1101-2]	AVG	
15	ABC-038	291,540	PRI /SEC /AVG NO.2 LPH LEVEL CONTROL	AVG / PRI / SEC	AI0674 [TT-FG1125]	AI0850 [LT-HD1102-2]	AVG	
16	ABC-039	291,540	PRI /SEC /AVG NO.3 LPH LEVEL CONTROL	AVG / PRI / SEC	AI0675 [TT-FG1092-B]	AI0851 [LT-HD1103-2]	AVG	
17	ABC-050	521,628	PRI /SEC /AVG HP TURBINE BYPASS STEAM PRESSURE CONTROL	AVG / PRI / SEC	AI0416 [PT-FG1141-A]	AI0512 [MIF-CL1061-B]	AVG	
18	ABC-050	151,217	AVERAGE SELECT CIRCUIT HP TURBINE BYPASS STEAM PRESSURE CONTROL	AVG	SI1814	SI1813	AVG	

#	Sheet	Place (x,y)	Drawing text	Possible modes	Input 1 address [IO tag]	Input 2 address [IO tag]	Sim now	DCS default (write)
19	ABC-050	151,172	AVERAGE SELECT CIRCUIT HP TURBINE BYPASS STEAM PRESSURE CONTROL	AVG	SI1817	SI1816	AVG	
20	ABC-051	631,625	PRI /SEC /AVG HP TURBINE BYPASS STEAM SPRAY WATER TEMP. CONTROL	AVG / PRI / SEC	AI0865 [TT-MS1022-1]	AI0897 [TT-MS1022-2]	AVG	
21	ABC-052	459,606	PRI /SEC /AVG LP TURBINE BYPASS STEAM CONTROL	AVG / PRI / SEC	AI0421 [TCS-AI1005]	AI0513 [FT-FA1081]	AVG	
22	ABC-052	763,621	PRI /SEC /AVG LP TURBINE BYPASS STEAM CONTROL	AVG / PRI / SEC	AI0864 [TT-HR1012-1]	AI0896 [TT-HR1012-2]	AVG	
23	ABC-052	198,585	AVERAGE LP TURBINE BYPASS STEAM CONTROL	AVG	AI0883 [PT-ST1001-1]	AI0884 [PT-ST1001-2]	AVG	
24	ABC-053	554,579	PRI /SEC /AVG TURBINE OIL COOLER OUTLET OIL TEMP. CONTROL	AVG / PRI / SEC	AI0452	AI0516	AVG	
25	ABC-002	733,555	PRI /SEC /AVG COMBUSTION AIR CONTROL	AVG / PRI / SEC	SIG.AB B.0671	SIG.AB B.0701	AVG	
26	ABC-003A	560,612	PRI /SEC /AVG A-START UP BURNER CONTROL (FUEL / AIR)	AVG / PRI / SEC	AI0657 [TT-FA1086]	AI0673 [TT-FA1087]	AVG	

F. TOTALIZERS (SUMA blocks, 10)

Time base of the input (the simulator integrates per HOUR: a flow of 36 T/H adds 0.01 T every second). Write the unit of the total and if the total resets (and at what value).

#	Sheet	Tag	Function block (MDL)	Input signal address [IO tag]	Description	Sim now	DCS input time base (write)	DCS unit of total (write)	Reset / limit (write)
1	ABC-003B	FIQDO1043B	S1-MDL174	AI0546 [FT-DO1043-B]	#1 Boiler start up burner-B supply line oil flow	per hour			
2	ABC-003C	FIQDO1043C	S1-MDL175	AI0517 [FT-DO1043-C]	#1 Boiler start up burner-C supply line oil flow	per hour			
3	ABC-003D	FIQDO1043D	S1-MDL176	AI0548 [FT-DO1043-D]	#1 Boiler start up burner-D supply line oil flow	per hour			
4	ABC-005	LIME-SUM	S1-MDL196	AI0151 [ST-M1311]	#1 Limestone feeder speed	per hour			
5	ABC-010	FIQFW10071	S1-MDL181	AI0592 [TT-FW1160] AI0256 [FT-FW1007-1]	#1 Boiler economizer #1 inlet water temperature	per hour			
6	ABC-010	FIQFW10072	S1-MDL182	AI0384 [FT-FW1007-2] AI0592 [TT-FW1160]	#1 Boiler feed water flow (2)	per hour			
7	ABC-032	FIQCD1121	S2-MDL086	AI0436 [FT-CD1121]	#1 Condensate polisher inlet water flow	per hour			
8	ABC-032	FIQCD1120	S2-MDL087	AI0437 [FT-CD1120]	#1 Condensate water flow	per hour			
9	ABC-001C	FIQMS1031	S1-MDL184	AI0131 [PT-MS1002] AI0129 [FT-MS1031]	#1 Boiler finishing superheater outlet main steam pressure	per hour			
10	ABC-003A	FIQDO1043A	S1-MDL173	AI0515 [FT-DO1043-A]	#1 Boiler start up burner-A supply line oil flow	per hour			